Thermodynamic description of the Pb–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Pb–Zn binary system as calculated with pycalphad from the SGTE dataset PbZn-98Jan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 + L2 → (Zn), lies at 420.0 °C.

1. Introduction

The Pb–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of T. Jantzen, P.J. Spencer, Calphad, 22, 417-34(1998). The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Pb reproduced by the dataset is 327.5 °C. For Zn, the liquidus meets the axis at 419.5 °C.

2. Phase diagram

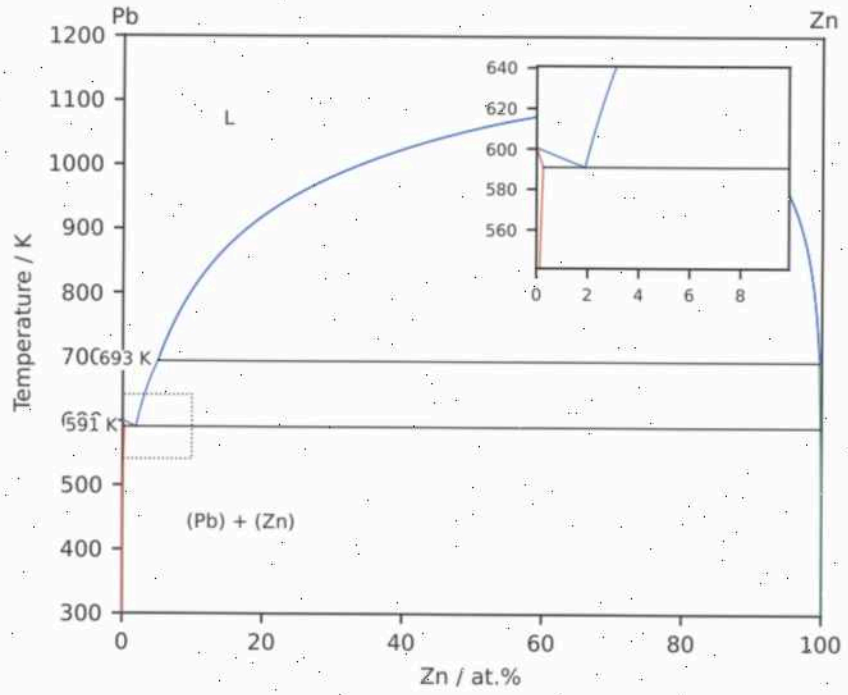


Fig. 1. Calculated Pb–Zn phase diagram (inset: enlarged view near the lowest invariant).

We obtain 420.0 °C for the syntectic reaction (L1 + L2 → (Zn)), where (Zn) has 99.6 at.% Zn. The dataset places a eutectic (L → (Pb) + (Zn)) at 317.8 °C; the composition of L is 1.9 at.% Zn.

Table 1. Invariant reactions in the Pb–Zn system (calculated).

Reaction	Type	T (°C)	at.% Zn, 1	at.% Zn, 2	at.% Zn, 3
L1 + L2 → (Zn)	syntectic	420.0	5.0	99.7	99.6
L → (Pb) + (Zn)	eutectic	317.8	1.9	0.3	99.8

3. Thermodynamic properties

At 1150 K the integral mixing enthalpy of the liquid has its maximum, 8486 J/mol, at 60.0 at.% Zn.

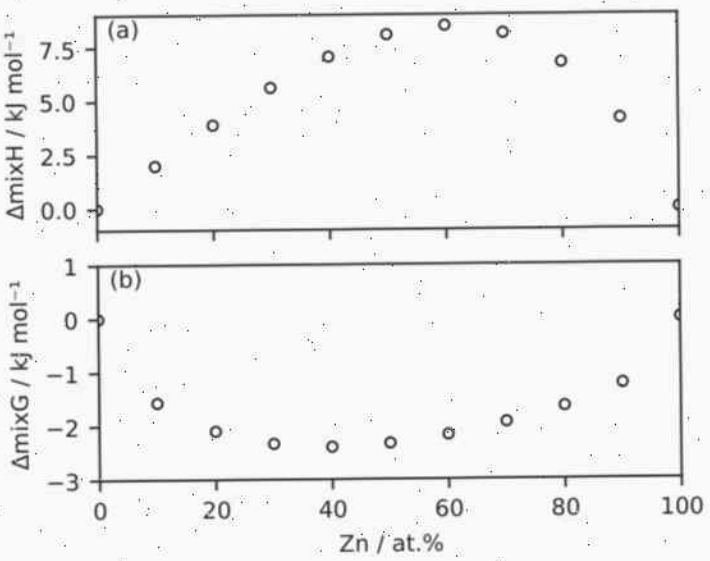


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1150 K.

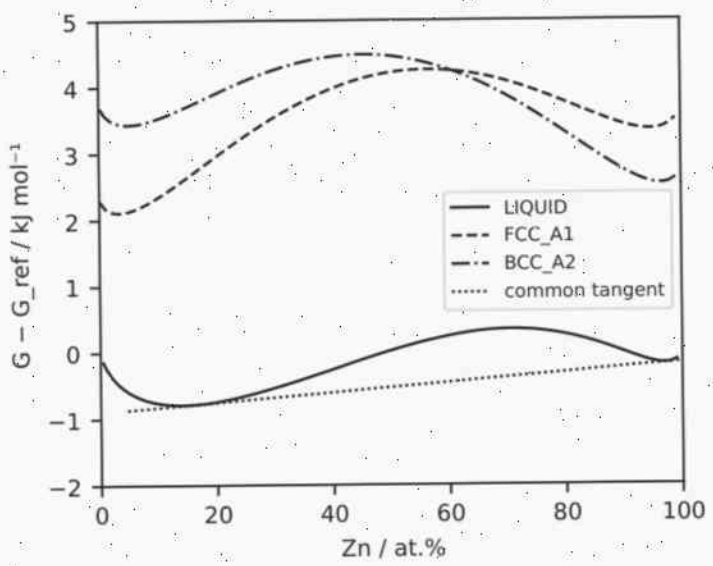


Fig. 3. Calculated Gibbs energy curves at 890 K.

Table 2. Redlich–Kister parameters of the liquid (Pb–Zn).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	32263.4	-13.10176	0.0
L1	-14909.2	10.92958	0.0
L2	2829.4	0.0	0.0

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.